Lec 2

ANN Model

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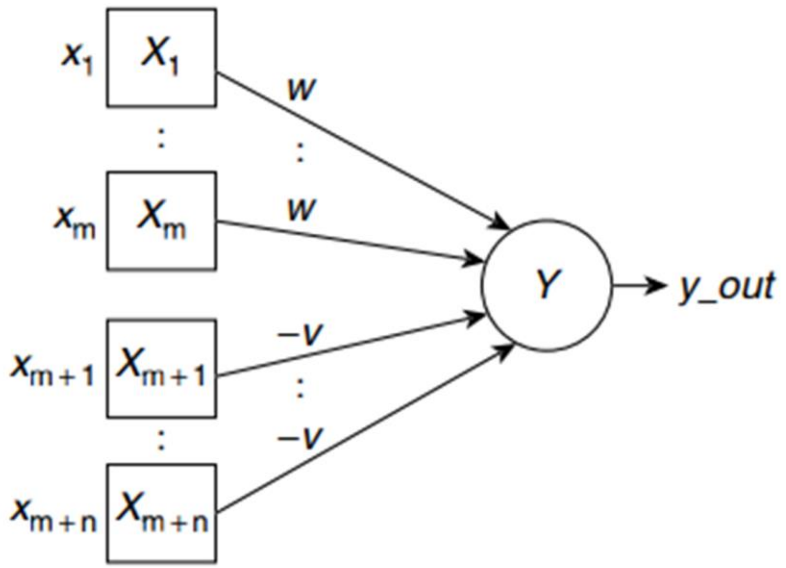
McCulloch Neural Network

- The earliest artificial neural model was proposed by McCulloch and Pitts in 1943. Fig. depicts its structure.
- *It consists of a number of input units connected to a single output unit. The interconnecting links are unidirectional.*
- *There are two kinds of inputs, namely*
  - *excitatory inputs*
  - *inhibitory inputs.*



- 1. There are two kinds of input units, excitatory, and inhibitory. In Fig the excitatory inputs are shown as inputs $X_1, \ldots, X_m$ and the inhibitory inputs are $X_{m+1}, \ldots, X_{m+n}$.
  - The excitatory inputs are connected to the output unit through **positively weighted links.**
  - Inhibitory inputs have **negative weights** on their connecting paths to the output unit.
- 2. All excitatory weights have the same positive magnitude w and all inhibitory weights have the same negative magnitude -v.

- 3. The activation $y\_out = f(y\_in)$ is binary, i.e., either 1 (in case the neuron fires), or 0 (in case the neuron does not fire).
- 4. The activation function is a binary step function. It is 1 if the net input ***y_in*** is greater than or equal to a given threshold value, and 0 otherwise.

$$y\_out = f(y\_in) = \begin{cases} 1, & if\ y\_in \geq \theta, \\ 0, & otherwise. \end{cases}$$

- 5. The inhibition is absolute. A single inhibitory input should prevent the neuron from firing irrespective of the number of active excitatory inputs.

For inhibition to be absolute, the threshold with the activation function should satisfy the following condition:

$$\theta > nw - p$$

The output will fire if it receives say "k" or more excitatory inputs but no inhibitory inputs, where

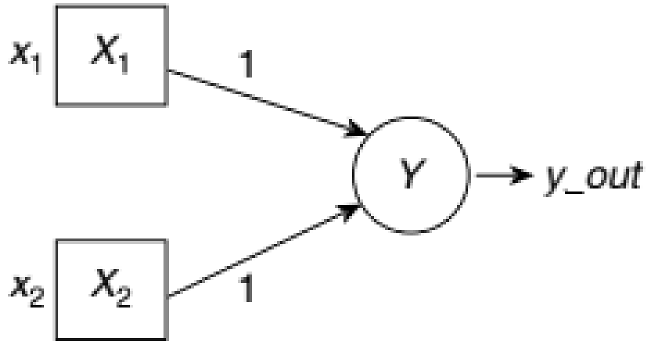
$$kw \geq \theta > (k-1)w$$

Implementation of logical AND with McCulloch-Pitts neural model

- select the appropriate number of inputs, the inter connection weights and the appropriate activation function.

(a) Truth Table

x_1	x_2	x_1 AND x_2
0	0	0
0	1	0
1	0	0
1	1	1



(b) Neural structure

$$y\_in = x_1 + x_2$$

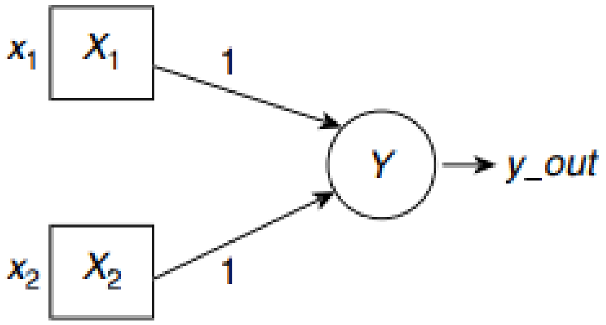
$$y\_out = f(y\_in) = \begin{cases} 1, & \text{if } y\_in \geq 2 \\ 0, & \text{otherwise.} \end{cases}$$

(c) Activation function

Implementation of logical OR with McCulloch-Pitts neural model

(a) Truth Table

x_1	x_2	x_1 OR x_2
0	0	0
0	1	1
1	0	1
1	1	1



(b) Neural structure

$$y\_in = x_1 + x_2$$

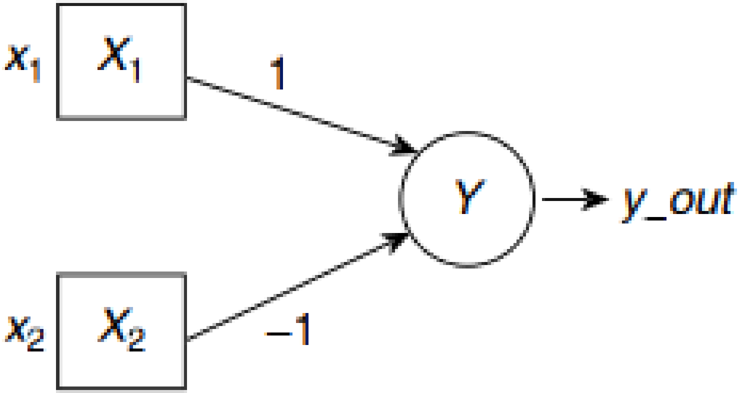
$$y\_out = f(y\_in) = \begin{cases} 1, & \text{if } y\_in \geq 1, \\ 0, & \text{otherwise.} \end{cases}$$

(c) Activation function

Implementation of logical AND-NOT with McCulloch-Pitts neural model

(a) Truth Table

x_1	x_2	x_1 AND (NOT x_2)
0	0	0
0	1	0
1	0	1
1	1	0



(b) Neural structure

$$y\_in = x_1 - x_2$$

$$y\_out = f(y\_in) = \begin{cases} 1, & \text{if } y\_in \geq 1, \\ 0, & \text{otherwise.} \end{cases}$$

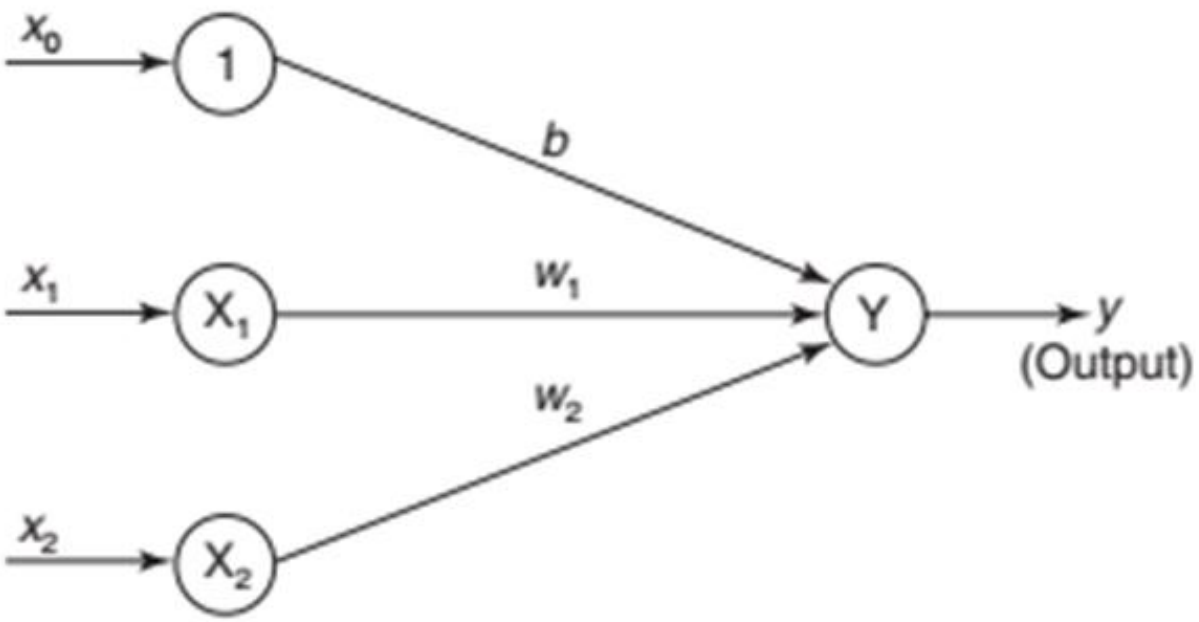
(c) Activation function

LINEAR SEPARABILITY

- An ANN does not give an exact solution for a nonlinear problem. However, it provides possible approximate solutions to nonlinear problems.
- Linear separability is the concept wherein the separation of the input space into regions is **based on** whether the **network response** is positive or negative.
- A decision line is drawn to separate positive and negative responses.
- The decision line may also be called as the decision-making line or decision-support line or linear-separable line.
- The necessity of the linear separability concept was felt to classify the patterns based upon their output responses.

- Generally the net input calculated to the output unit is given as

$$y_{in} = b + \sum_{i=1}^{n} x_i w_i$$



- If a bipolar step activation function is used over the calculated net input (*yin*) then the value of the function is 1 for a positive net input and –1 for a negative net input.
- Also, it is clear that there exists a boundary between the regions where *yin* > 0 and *yin* < 0.
- This region may be called as *decision boundary* and can be determined by the relation

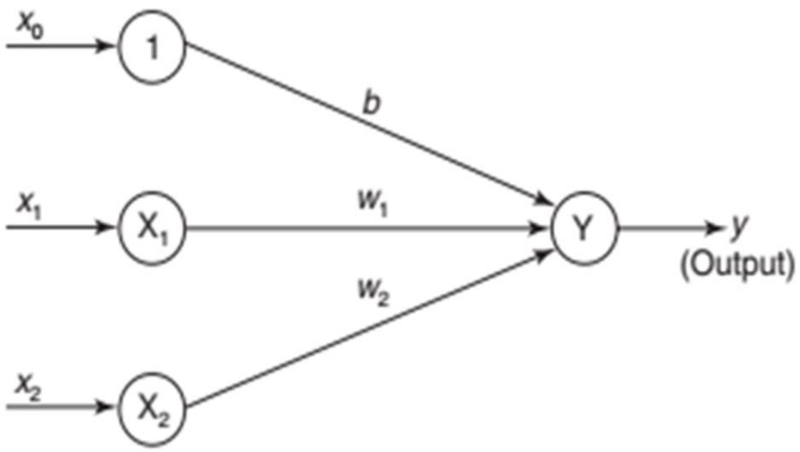
$$b+\sum_{i=1}^{n} x_i w_i = 0$$

$$b+\sum_{i=1}^{n} x_i w_i = 0$$

On the basis of the number of input units in the network, the above equation may represent **a line,** a plane or a hyperplane.

If there exist weights (with bias) for which the training input vectors having positive (correct) response, +1, lie on one side of the decision boundary and all the other vectors having negative (incorrect) response, -1, lie on the other side of the decision boundary then we can conclude the problem is linearly separable.

- Consider a single-layer network as shown in Figure with bias included. The net input for the network shown in Figure is given as



$$y_{in} = b + x_1 w_1 + x_2 w_2$$

- The separating line for which the boundary lies between the values *x*1 and *x*2, so that the net gives a positive response on one side and negative response on other side, is given as

$$b + x_1 w_1 + x_2 w_2 = 0$$

- If weight *w*2 is not equal to 0 then we get

$$x_2 = -\frac{w_1}{w_2}x_1 - \frac{b}{w_2}$$

- Thus, the requirement for the positive response of the net is

$$b + x_1 w_1 + x_2 w_2 > 0$$

- During training process, the values of $w1$, $w2$ and b are determined so that the net will produce a positive (correct) response for the training data.
- If on the other hand, threshold value is being used, then the condition for obtaining the positive response from output unit is

$$\text{Net input received} > \theta\,(\text{threshold})$$

$$y_{in} > \theta$$

$$x_1 w_1 + x_2 w_2 > \theta$$

- The separating line equation will then be $x_1 w_1 + x_2 w_2 = \theta$

$$x_2 = -\frac{w_1}{w_2} x_1 + \frac{\theta}{w_2} \quad (\text{with } w_2 \neq 0)$$

- During training process, the values of $w1$ and $w2$ have to be determined, so that the net will have a correct response to the training data. For this correct response, the line passes close through the origin.

- calculate the net input to the output neuron
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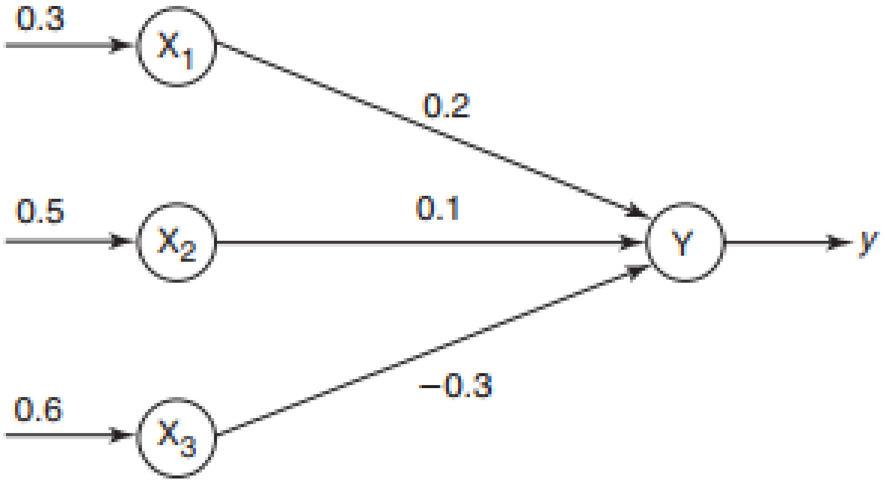

HEBB Network